Closest-Feasible Replication of Chen, Han, and Pan (2021): Tables III and IV

Jeffrey (Jiahui) Chen

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Code and aggregate outputs (no restricted data):
`https://github.com/Jeffrey-Chen-arch/chen-han-pan-2021-replication`

1 Objective, data, and method

This report documents a closest-feasible replication of Tables III and IV of Chen, Han, and Pan, "Sentiment Trading and Hedge Fund Returns" (*Journal of Finance*, 2021). The published code provides final-table logic but not the main sample construction, so the aim is to reproduce the published results as closely as the available specifications, the authors' SAS logic, and the public factor inputs allow, and to document the remaining differences. All "Paper" values below are the published values in CHP (2021).

Data. Fund data are Lipper/TASS via WRDS. Applying the paper's screens (U.S. equity-oriented categories, USD monthly net-of-fee returns, monthly-or-better redemption, AUM $\geq$ \$5m, drop the first 12 months, $\geq$ 30 return observations, de-duplicate) yields **2,589 funds** over 1994–2018, versus 4,073 in the paper. Public factors are from Kenneth French, David Hsieh (PTFS), FRED, Pastor–Stambaugh, and Baker–Wurgler.

Method. Each month a fund's sentiment beta is its loading on the Baker–Wurgler sentiment-*changes* index in a 36-month rolling regression of excess returns on that index and the Fung–Hsieh factors, momentum, liquidity, inflation, and the default spread (the paper's eq. (1)). Funds are sorted into ten equal-weighted deciles held one month; alphas use the Fung–Hsieh seven factors, momentum, and liquidity, excluding inflation and the default spread (footnote 15), with Newey–West(2) errors. The sentiment-changes index is the first principal component of the orthogonalized proxy changes; term/credit factors are proxied from FRED because the paper's Barclays return series are not public.

2 Main results: replication vs. paper

	Excess return (%/mo.)		Alpha (%/mo.)	
Decile	Replication	Paper	Replication	Paper
1 (Low)	0.26	0.27	0.01	−0.08
2	0.20	0.25	0.05	0.07
3	0.24	0.34	0.23	0.14
4	0.26	0.30	0.28	0.14
5	0.25	0.25	0.25	0.04
6	0.28	0.25	0.31	0.10
7	0.26	0.30	0.16	0.12
8	0.28	0.34	0.23	0.21
9	0.36	0.38	0.34	0.26
10 (High)	0.56	0.58	0.43	0.51
Spread (10−1)	0.30 (t 2.43)	0.31 (t 3.16)	0.43 (t 1.64)	0.59 (t 3.55)

Table 1: Table III — decile portfolios sorted on sentiment beta. Sentiment-beta spread: replication 3.56 (2.22 after rescaling to the paper's Table II sentiment SD) vs. paper 2.03; PCA scale is arbitrary, so the rescaled value is the comparable one and rescaling does not change rankings, returns, or alphas. The full table — the sentiment beta and the excess- and alpha-t-statistics for every decile — is in the results workbook.

Table IV model (coef. on sentiment beta, t)	Replication	Paper
Excess return, univariate	0.106 (2.42)	0.17 (3.03)
Excess return, multivariate	0.091 (1.79)	0.16 (2.84)
Alpha, univariate	0.127 (1.71)	0.14 (3.31)
Alpha, multivariate	0.062 (0.75)	0.12 (3.16)

Table 2: Table IV — Fama–MacBeth coefficient on sentiment beta (t-statistic in parentheses). The multivariate models control for the paper's fund characteristics (lagged log size and age, management and incentive fees, high-water-mark and lockup indicators, and notice period, plus style dummies); the results workbook additionally reports each model's adjusted R^2 and average monthly cross-section.

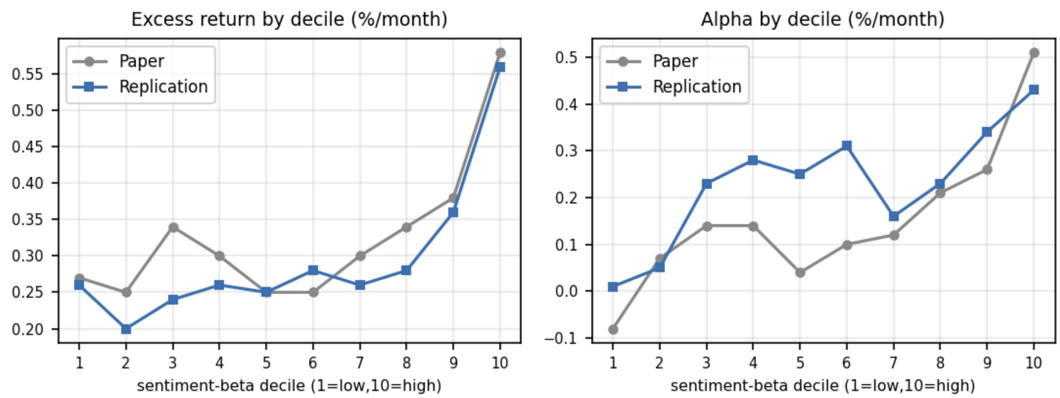

Figure 1: Decile pattern, replication vs. paper. Both the excess return and the alpha rise with the sentiment beta. Several economic magnitudes are close, especially the Table III excess spread and the Table-II-scaled beta spread; statistical significance is weaker than in the paper.

3 Structured specification grid and main alternatives

Rather than tune a single path, I evaluated a pre-specified, admissible grid and scored each combination on closeness to the published tables, keeping main-eligible and diagnostic specifications separate.

Dimension	Variants examined	Outcome / choice
Sample screens	paper-strict (2,270); keep-missing redemption (2,589); +lenient AUM (2,748); all-categories, IA (2,625); alternative filters, IA (1,793); fully relaxed (3,169); company-cluster identifier (1,396)	No admissible screen reaches 4,073; chose keep-missing (2,589).
Sentiment construction	M1 correlation-PCA; M2 covariance-PCA; M3 orthogonalize-changes-then-PCA; M4 level-PC difference; M5 $\Delta\text{SENT}^{\perp}$; M6 ΔSENT; Michigan index	Classic M1 is weak (alpha $\approx$ 0.08–0.22); M3 is the strongest paper-faithful construction (0.43).
Factor proxy	F0 professor-guided FRED level term-spread; F1 Hsieh yield-change	F0 (0.43); F1 (0.29); both fall short of 0.59.
Horizon / estimation	next-month (main); skip-month, two-step beta, 3- and 6-month holding (IA)	Main = next-month; IA variants scored only against their own IA targets.

Table 3: What I tried (Table III alpha spread shown where relevant).

4 Robustness and integrity checks

Check	Effect on alpha spread	Decision
1/99 winsorization of fund returns	0.43 → 0.40	The paper winsorizes the Table I summary statistics; tested as a robustness check; does not close the gap.
Fung–Hsieh yield-change factor (F1)	0.43 → 0.29	Closer to Hsieh's definition, but not the exact paper input: the paper uses Barclays return-based factor-mimicking portfolios, which are unavailable.
Drop inflation/default from beta reg.	0.47 → 0.64 (proxy spec)	**Rejected**: the paper's eq. (1) includes inflation and the default spread; adopting this would deviate from the paper.
No-cache cold re-run of main spec	identical (≤ 0.0004)	Results reproduce from raw inputs.

Table 4: Sensitivity checks on the primary specification.

Among specifications that follow the paper's main design (next-month horizon, Baker–Wurgler sentiment, a main-paper sample, and paper/professor-guided factors), this one is the closest on the return spreads while keeping every regression exactly as the paper specifies. Over 1997–2018 the monthly risk-adjusted high-minus-low spread is positive in 61.7% of months, so the pattern is not driven by a few isolated months, though it is weaker than the paper's Figure 1 (positive in more than two-thirds of months).

5 Difficulties and discrepancy audit

Remaining differences appear most consistent with unrecoverable inputs and sample-construction details rather than a single identifiable implementation issue; the items below are stated as possible sources.

Open item	Status / note
Exact 4,073-fund TASS universe	No admissible screen or identifier reaches 4,073 (ceiling 3,169; the WRDS `productreference` is the finest unit); likely contributes materially to the weaker t-statistics, together with sentiment and factor-input differences.
Exact Baker–Wurgler sentiment file	Results are sensitive to the construction (classic correlation-PCA weak, an alternative stronger); the authors' exact series is not recoverable.
Barclays factor returns	Bond/credit factors are built from non-public Barclays returns; proxied from FRED.
Table IV alpha construction	SAS full-sample alpha and the paper-text rolling alpha are both tested.

Table 5: Open items (possible sources of the remaining gap).

6 Conclusion

This is a closest-feasible replication: the main economic magnitude is substantially recovered — the excess-return spread (0.30% vs. 0.31%) and the rescaled sentiment-beta spread (2.22 vs. 2.03) — while the full statistical strength is not (alpha spread 0.43%, $t = 1.64$, vs. 0.59%, $t = 3.55$). Every regression was verified against the paper, and a specification that would have raised the alpha but deviates from the paper's eq. (1) was rejected. The remaining gap is most consistent with unrecoverable inputs — the exact fund universe, the Baker–Wurgler sentiment file, and the Barclays factor returns — rather than a fixable specification choice.